XRPL Validator — Daily Engineering Report · 2026-03-25

Period: 2026-03-25 00:00 to 24:00 US Eastern (2026-03-25 04:00 to 2026-03-26 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 2 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix: pinned-ledger bulk sync, disable background DB writers** (`a6ec576`, xrpl-validator)
- Bulk sync now pins all workers to a single ledger_index for consistent state - Split keyspace across 3 HTTP endpoints (1 worker each, throttled) - Disabled background fetcher threads that wrote "validated" ledger data into the pinned-ledger DB, corrupting the state snapshot - Disabled close_ledger deletes that removed entries expecting re-fetch - Added incremental sync module with sweep resume capability - Added …
- **perf: SHAMap memory optimization — 42GB to 13GB RSS** (`423028c`, xrpl-validator)
- Eliminate clone-on-insert: insert/delete now mutate in-place instead of deep-cloning the entire 18.7M-node tree on every call - Hash-only leaf nodes: tree stores pre-computed hashes, not full serialized ledger objects (saves ~6GB of duplicated data) - Hash caching on inner nodes: root_hash() only recomputes dirty paths instead of all 18.7M+ nodes - Fixed-size [Option; 16] array replaces Vec in …

Independence program

No independence record for UTC day 2026-03-25; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
03-26 01:47	—	xrpl-validator	`a6ec576`	fix: pinned-ledger bulk sync, disable background DB writers
03-26 02:43	—	xrpl-validator	`423028c`	perf: SHAMap memory optimization — 42GB to 13GB RSS

Times in UTC. A commit appears on the day it was committed to main or merged into it.